AAE530 COMPUTATIONAL COMBUSTION AND PROPULSION

(3 credits)

COURSE INFORMATION/DESCRIPTION:

The course covers the following topics:

- Fundamentals of thermochemistry
- Chemical equilibrium and its calculation
- Chemical kinetics and auto-ignition.
- Laminar non-premixed flames and computation of an opposed jet flame.
- Laminar premixed flames and calculation of the laminar flame speed.
- Models for turbulent combustion (the flamelet model and the transported probability density function model).
- Turbulent non-premixed combustion and the modeling and simulation of a turbulent free jet flame.
- Turbulent partial premixed combustion and the modeling and simulation of a turbulent lifted jet flame.
- Computational propulsion and the modeling of a model rocket combustor
- [optional] Advanced topics on data-driven modeling and machine learning.

The course consists of lectures and computer projects.

LEARNING RESOURCES AND REFERNCE TEXT

PURDUE BRIGHTSPACE is used to organize the course contents such as announcements, course notes, assignments, report submission, and grades. Please check Brightspace frequently to get the latest update.

REQUIRED TEXTBOOK none but a modern text on combustion (e.g., Turns) and turbulence (e.g., Pope) are recommended.

ASSIGNMENTS:

Quizzes

During lectures, in-class quizzes will be administered occasionally. These quizzes are intended to reinforce comprehension, foster class participation, and serve as both a teaching and learning evaluation tool.

Project

There are bi-weekly course project assignments. A project report is expected to be submitted and graded.

COURSE LEARNING OBJECTIVES:

- Calculate chemical equilibrium.
- Calculate rate of chemical reaction and compute auto-ignition process.
- Compute opposed jet flames and understand characteristics of non-premixed combustion;
- Compute laminar flame speed and understand characteristics of premixed combustion.
- Acquire basic understanding of turbulent combustion problems and the advanced modeling approaches.

- Choose appropriate models for turbulent combustion problems and perform turbulent combustion simulations.

EVALUATION:

Course Grade

The final grades are computed based on the project evaluation scores.

Grade Scale

The course uses the course averages in the recent semesters as the reference scale for assigning the final grades.

Grade	Justification
A, A±	Significantly above average*
B, B±	Near and above average
C, C±	Slightly below average
D, D±	Significantly Below average
F	Extremely poor performance